Question

A crystalline compound with the chemical formula $\mathbf{AB}$ belongs to the tetragonal crystal system and has a density of $10.27 \mathrm{g/cm^3}$. Both $\mathbf{A}$ and $\mathbf{B}$ are 4-coordinated, and their coordination configurations are different. There is only one type of $\mathbf{A}-\mathbf{B}$ bond length in the crystal, which is 231.0 pm. The structure can be regarded as an infinite chain structure of $\mathbf{AB}_2$. The chains in the same xy plane are parallel to each other. The chains can be regarded as ribbon-like chains extending in the a or b direction, with a certain width in the c direction. The chains in adjacent xy planes are perpendicular to each other and share vertices. The shortest $\mathbf{A}-\mathbf{A}$ distance is 347.0 pm. There are the following statements:

1. The lattice type of the crystal is tI.

2. All $\mathbf{B}$ atoms in the crystal have the same spatial environment.

3. The fourth-period element in the same group as $\mathbf{A}$ exists in nature as a simple substance.

Then, all the correct options are:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 1,2

F. 1,3

G. 2,3

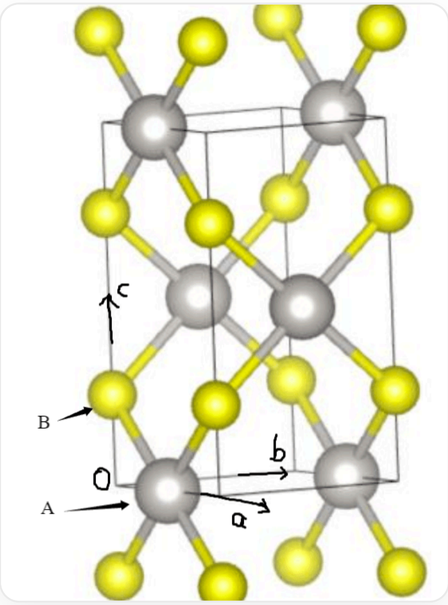
H. 1,2,3

Answer

Correct Answer: D

Detailed Explanation

From four coordination, it can be deduced that one atom should be square planar coordinated (forming $\mathbf{AB}_4$ edge-sharing connections to produce $\mathbf{AB}_2$), and the other atom should be tetrahedral. The unit cell is shown in the figure:



The image shows the unit cell of the compound, which is a tetragonal unit cell. The gray spheres represent A atoms with atomic coordinates (0.5,0,0) and (0,0.5,0.5). The yellow spheres represent B atoms with atomic coordinates (0,0,0.25) and (0,0,0.75). In addition, the image also shows parts of B atoms in the unit cells above and below the current unit cell

From the figure, it is easy to see that the lattice form of the crystal is tP.

CHECKPOINT 1 PTS

The crystal lattice form is tP

There are two orientations of $\mathbf{A}$ atoms around the $\mathbf{B}$ atom.

CHECKPOINT 1 PTS

There are two spatial environments for $\mathbf{B}$ atoms in the crystal

The unit cell parameters can be obtained through calculation:

$$a=347\mathrm{pm}, c=610\mathrm{pm}$$

Thus, we have the equation:

$$\rho=\frac{ZM}{N_AV}=\frac{2M}{N_A*347*347*610*10^{-30}\mathrm{cm}^3}=10.27\mathrm{g/cm}^3$$

Solving for $M=227.1\mathrm{g/mol}$, which leads to PtS, $\mathbf{A}$ is Pt, belonging to group VIII. The fourth-period elements in the same group are $\mathrm{Fe, Co, Ni}$. Among them, Fe exists as native metal in meteorites in nature.

CHECKPOINT 1 PTS

The fourth-period element Fe in the same group as $\mathbf{A}$ exists as native metal in meteorites in nature

Therefore, choose option D